From Local Repulsion to Global Geometry: Large-Scale Tests of Geometric ETH

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Exactly degenerate quantum manifolds have no internal level spacings, so the usual spectral diagnosis of quantum chaos is silent. We test a geometric replacement: the statistics of the non-Abelian Berry curvature after normalization by the quantum metric. For a bosonic fractional-quantum-Hall zero-mode manifold, we generate 20000 physical tangent pairs, reserve a disjoint 4000-matrix test set, and compare it with 10000 Haar–Jacobi and 10000 covariance-deformed random matrices. Local adjacent-gap statistics agree at the 10^{-3} scale, whereas the global density is strongly non-Haar. A covariance model learned only from tangent-channel row spaces lowers the held-out density error from 0.2910 to 0.1237, an improvement of 57.5%. An independent root-space calculation of 8750 matrices extends the active rank to 800. When the rank exceeds the one-polarization capacity, the curvature develops exactly $r - M$ atoms at each of $\lambda = \pm 1$; at the largest point these comprise 30.0% of the spectrum, while the remaining continuous spectrum retains Jacobi repulsion and an interior density error 0.0523. These results support a hierarchical geometric eigenstate thermalization hypothesis: local curvature correlations become universal before global density and Grassmannian frame statistics, and exact geometric modes can coexist with a chaotic complementary sector.

I. INTRODUCTION

Random-matrix level statistics and the eigenstate thermalization hypothesis (ETH) are among the sharpest operational signatures of quantum chaos [1–4]. Both are organized by energy: levels repel after resolving symmetry sectors, and matrix elements fluctuate as a smooth function of mean energy and frequency. An exactly degenerate multiplet defeats this organization at the outset. Every internal spacing vanishes, the spectral form factor of the multiplet is trivial, and an "ETH ansatz inside the band" has no frequency variable on which to depend.

The state bundle over coupling space provides a different arena. For an isolated D-dimensional projector $P(\lambda)$, adiabatic transport is governed by the Wilczek–Zee connection and its matrix-valued curvature [5–7]. The same derivatives define the quantum metric [8, 9]. Chen, Colin-Ellerin, Mamroud, and Papadodimas recently proposed the non-Abelian Berry curvature as an intrinsic chaos diagnostic for exactly degenerate BPS microstates [10]. Their central contrast is geometric rather than spectral: smooth horizonless sectors have structured, often vanishing curvature, while supersymmetric black-hole microstates exhibit random-matrix-like curvature. The curvature also probes topology through Chern numbers over moduli space.

Here we ask what survives of this idea in a controlled condensed-matter setting with no supersymmetric cohomology. Our manifold consists of exact zero modes of a frustration-free fractional-quantum-Hall parent Hamiltonian [11–14]. The degeneracy is algebraic and topological, but it is not a BPS index and is not protected by SUSY cohomology. This distinction is useful: it isolates the geometry of an exactly degenerate projector from the specifically supersymmetric mechanisms of Ref. [10].

The object we study is the metric-normalized curvature

$$\Omega = \Gamma^{-1/2} F_{vw} \Gamma^{-1/2}, \qquad \Gamma = X_v X_v^\dagger + X_w X_w^\dagger, \quad (1)$$

restricted to the support of Γ. The matrices X_v and X_w are tangent channels from the degenerate manifold to its gapped complement. Normalization removes the nonuniversal strength of the two perturbations and leaves a compression of a fixed signature form by a row isometry. If that row space is Haar distributed, the spectrum is exactly a finite Jacobi unitary ensemble [15, 16]. This gives a parameter-free null model at the observed active rank and complement dimension.

Our results establish three scales of geometric randomness. First, in a 20000-matrix physical ensemble, the adjacent-gap distribution of Ω is indistinguishable at current resolution from the Haar–Jacobi law, although the one-point density is not. Second, the leading global deformation is explained by a covariance-deformed row-space ensemble trained and validated without access to the final test split. Third, an increasing-rank root-space sequence reveals an exact capacity transition. When the active rank r exceeds the dimension M of either polarization of the channel space, dimension counting forces $r - M$ eigenvalues at each of $+1$ and -1. The continuous complement nevertheless retains random-matrix local correlations.

This last observation changes the interpretation of exact modes. An exact spectral component need not signal an integrable curvature matrix. Instead, the curvature decomposes into a geometry-enforced atomic sector and a chaotic continuous sector. The result is a concrete form of *partial geometric ergodicity*: different observables of the same row-space ensemble forget microscopic structure at parametrically different rates.

The paper is organized as follows. Section II derives the signature-compression form, its Jacobi law, the atom theorem, and the covariance-deformed geometric-ETH ansatz. Sections III and IV present the high-statistics physical test and the held-out covariance mechanism. Section V crosses the exact atom boundary through $D = 800$. Section VI compares local repulsion, number variance, rigidity, and the connected form factor. Appendix A records the ensembles, matrix-level inference, sensitivity analysis, and reproducibility gates.

II. GEOMETRY AND RANDOM-MATRIX ENSEMBLES

A. From projector derivatives to a row space

Let $H(\lambda)$ have an isolated D-dimensional eigenspace at energy E_0, with spectral projector P, complement $Q = 1 - P$, and reduced resolvent

$$R = Q\big[Q(H - E_0)Q\big]^{-1}Q. \tag{2}$$

For a tangent $V_\mu = \partial_\mu H$, define the channel

$$X_\mu = PV_\mu R. \tag{3}$$

In a P-basis and a basis of the coupled complement, X_μ is $D \times M$. Kato perturbation theory gives the pulled-back quantum geometric tensor; for two real tangent directions v, w, its antisymmetric part is

$$F_{vw} = i\left(X_v X_w^\dagger - X_w X_v^\dagger\right). \tag{4}$$

The positive form entering Eq. (1) is the sum of the two channel metrics.

Introduce the doubled channel matrix and a fixed Hermitian signature form,

$$B = \begin{pmatrix} X_v & X_w \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & iI_M \\ -iI_M & 0 \end{pmatrix}. \tag{5}$$

On the $r = \operatorname{rank} B$ dimensional support of $\Gamma = BB^\dagger$, whitening produces

$$Y = \Gamma^{-1/2}B, \qquad YY^\dagger = I_r, \qquad \Omega = YJY^\dagger. \tag{6}$$

Thus normalized curvature depends only on an r-plane in $\mathbb{C}^{2M}$. Changing the basis of the active subspace conjugates Ω and leaves every spectral statistic invariant. The problem is consequently a probability problem on a complex Grassmannian rather than on the original Hilbert space.

B. Haar compression and the exact Jacobi law

If the row space of Y is Haar distributed, Eq. (6) is a random compression of a matrix with M eigenvalues $+1$ and M eigenvalues -1. For $r \le M$, the r eigenvalues have joint density

$$p(\lambda_1, \dots, \lambda_r) \propto \prod_{a<b} |\lambda_a - \lambda_b|^2 \times \prod_{a=1}^{r} (1 - \lambda_a^2)^{M-r}, \qquad -1 < \lambda_a < 1. \tag{7}$$

This is the complex Jacobi ensemble. It contains both the one-point confinement and the Vandermonde repulsion; matching only the density is therefore weaker than matching the full finite-(r, M) law.

The law is sampled independently in our calculation through the matrix-beta construction. Let $k = r$ and $A = GG^\dagger$, $B = HH^\dagger$, where G, H are independent $k \times M$ complex Gaussian matrices. The generalized eigenvalues t of $(A, A + B)$ map to $\lambda = 2t - 1$ and obey Eq. (7). This avoids using physical tangent data to manufacture the reference ensemble.

C. Exact boundary atoms

The regime $r > M$ is qualitatively different. Let $\mathcal{S} \subset \mathbb{C}^{2M}$ be the row space of Y, and let $\mathcal{H}_\pm$ be the M-dimensional ± 1 eigenspaces of J. The intersection inequality gives

$$\dim(\mathcal{S} \cap \mathcal{H}_\pm) \ge \dim \mathcal{S} + \dim \mathcal{H}_\pm - 2M = r - M. \tag{8}$$

Every vector in $\mathcal{S} \cap \mathcal{H}_\pm$ is an exact ± 1 eigenvector of the compression. For a generic r-plane the inequality is saturated. Therefore

$$a_+ = a_- = r - M, \qquad k = 2M - r, \tag{9}$$

where k is the dimension of the remaining continuous spectrum. After removing these levels by their algebraic labels, not by a numerical proximity cut, the interior again has a Jacobi density,

$$p_{\text{int}}(\lambda) \propto \prod_{a<b} |\lambda_a - \lambda_b|^2 \prod_a (1 - \lambda_a^2)^{r-M}. \tag{10}$$

Equations (9) and (10) are purely kinematic consequences of the signature compression. They do not assume chaos. Chaos is tested only in the distribution of the continuous complement.

D. Covariance-deformed geometric ETH

Haar row spaces are the strongest possible isotropy hypothesis. A weaker geometric ETH allows a deterministic channel covariance C:

$$Z = GC^{1/2}, \qquad Y_C = (ZZ^\dagger)^{-1/2}Z, \tag{11}$$

where G is an $r \times 2M$ complex Gaussian matrix. For $C \propto I$, Y_C is Haar. For nontrivial C, the ensemble retains local Gaussian mixing but has a biased mean row-space projector and a deformed one-point law.

This ansatz is the geometric analogue of separating a smooth ETH envelope from a random residual. It does not state that microscopic channels are independent Gaussian variables. Rather, it asks whether their spectral consequences are organized predominantly by a two-point covariance, with higher Grassmannian cumulants appearing as controlled residuals. The distinction between local repulsion and global density will test precisely this hierarchy.

III. HIGH-STATISTICS PHYSICAL CURVATURE LAW

A. Fractional topological zero-mode manifold

The physical anchor is a bosonic Kapit–Mueller lattice parent projected to its exactly flat lowest band [12]. At particle number $N = 3$ and flux number $n = 10$, the projected contact interaction has an isolated zero-mode manifold of dimension $D = 50$. The complement coupled by the local tangent family has dimension $M = 170$ per tangent direction. The zero modes are the lattice counterpart of bosonic Laughlin quasihole states and obey the cyclic $(1, 2)$ exclusion rule [11, 13].

We emphasize what is and is not protected. The parent is positive semidefinite and the manifold is its exact kernel at the sampled base point. No supersymmetry algebra, Witten index, or cohomology is invoked. The open external gap makes the projector and its Berry curvature well defined, while the exact degeneracy makes conventional internal level statistics unavailable.

The tangent directions are independently drawn mean-zero local potentials,

$$V[v] = \sum_{\boldsymbol{x}} v_{\boldsymbol{x}}\, \hat{n}_{\boldsymbol{x}}, \qquad \sum_{\boldsymbol{x}} v_{\boldsymbol{x}} = 0, \tag{12}$$

and normalized in the many-body Hilbert–Schmidt metric before channels are formed. We generate 20000 tangent pairs in eight independently seeded blocks. A fixed permutation divides them into training, validation, and test sets; the final 4000 matrices are untouched by covariance estimation and regularization choice.

B. Local universality without global Haar equality

Figure 1 compares the test ensemble with two independently generated references. The black curve is the exact finite-(r, M) Haar–Jacobi law. The orange curve is the covariance ensemble of Eq. (11), fit as described in Sec. IV. Bands are simultaneous confidence regions obtained by resampling the eight physical seed blocks or independent reference matrices.

The local adjacent-gap means are

$$\begin{aligned} &\langle r\rangle_{\text{phys}} = 0.599806, \\ &\langle r\rangle_{\text{Haar}} = 0.599395, \qquad \langle r\rangle_{C} = 0.599200. \end{aligned} \tag{13}$$

The full $P(r)$, shown later in Fig. 5, agrees across the three ensembles within simultaneous bands. This conclusion is stable under five bulk windows, two unfolding procedures, and three histogram resolutions.

Global equality fails decisively. The physical density extends farther toward the signature edges and is flatter than the Haar density. The held-out density L^1 distance is 0.2910. Higher even moments amplify the same mismatch: the Haar result progressively underestimates the physical tails from the second through the eighth moment. Consequently, "random-matrix-like Berry curvature" cannot mean equality of every spectral observable to one invariant ensemble. At this finite rank, the supported statement is local Jacobi universality in a globally anisotropic row-space distribution.

IV. HELD-OUT COVARIANCE MECHANISM

The natural invariant diagnostic of row-space anisotropy is the ensemble-mean ambient projector

$$\overline{\Pi} = \mathbb{E}\left[Y^{\dagger} Y\right]. \tag{14}$$

For Haar r-planes in $L = 2M$ dimensions, $\overline{\Pi}_{\text{Haar}} = (r/L) I_L$. The physical training ensemble instead has relative Frobenius anisotropy 0.867 and coordinate participation 0.625. Its row spaces are therefore not small fluctuations around an isotropic mean.

We estimate Eq. (14) from 1024 training row spaces and use it as the covariance C in Eq. (11). Because the empirical covariance is low rank, its eigenvalues are regularized by a floor relative to the largest eigenvalue. Five floors are preregistered. The floor 0.050 is selected using validation density and gap-ratio errors only; the test data play no role in this choice.

On the untouched test set, the covariance model lowers the density error from 0.2910 to 0.1237, a 57.5% reduction. It simultaneously preserves the local result in Eq. (13). The improvement is not an artifact of the plotting bandwidth: for every registered bandwidth, the covariance error is below the Haar error by a wide margin. Moments through eighth order show that the same covariance controls much of the tail enhancement, not only the central density.

The fit is not exact. The physical mean projector, frame overlaps, and entry fourth moment retain structure beyond a single covariance matrix. Moreover, the longest-range number variance remains measurably above both random ensembles. We therefore call the result *covariance-deformed geometric ETH*, not full Haar thermalization. Its content is an ordering of cumulants:

$$\begin{aligned} \text{local repulsion} \;&\prec\; \text{one-point covariance envelope} \\ &\prec\; \text{higher frame correlations,} \end{aligned} \tag{15}$$

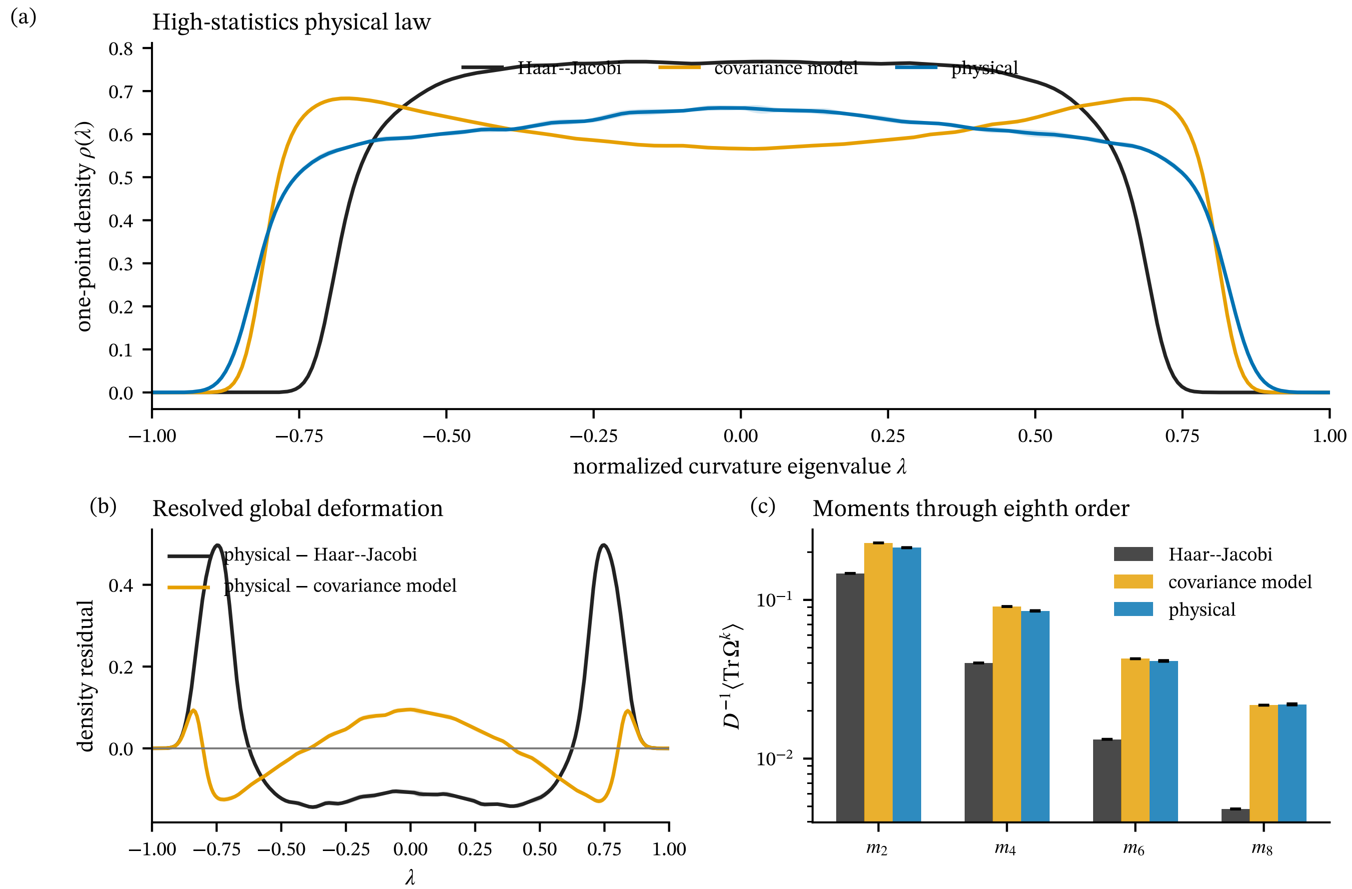

FIG. 1. **High-statistics physical curvature law.** (a) One-point density of the metric-normalized Berry curvature for the held-out physical test set, exact Haar–Jacobi reference, and covariance-deformed model. Shaded regions are simultaneous matrix- or seed-block-level confidence bands. (b) Density residuals expose the global failure of Haar isotropy and the smaller covariance residual. (c) Trace moments through eighth order; uncertainties resample matrices rather than pooled eigenvalues.

where $\prec$ denotes loss of microscopic information at a shorter geometric scale.

V. INCREASING RANK AND THE EXACT ATOM CROSSOVER

A. Exact root-response sequence

The physical lattice calculation provides high statistics at one active rank. To separate rank growth from the cost of diagonalizing the full parent Hamiltonian, we construct an independent sequence directly in the exact cyclic Laughlin root spaces. For $N = 3$ particles and n orbitals, the zero-mode count and one-tangent descendant capacity are

$$D(n) = \frac{n}{n-3}\binom{n-3}{3}, \qquad M(n) = 2n(n-3). \tag{16}$$

The two tangent matrices are independent real-symmetric one-body operators. Their exact monomial response blocks are assembled sparsely and then whitened by a thin QR factorization. This produces the same spectrum as Eq. (6) without forming an ill-conditioned inverse square root.

The registered sequence is summarized in Table I. It contains 8750 root-response matrices and reaches $D = 800$, nearly four times the previous largest-rank calculation in the project. Every case is divided into eight seed blocks, and every reference is an independent complex Wishart–Jacobi ensemble.

B. Continuous chaos plus exact geometry

Figure 3 displays the full qualitative transition. For $D \le M$, all curvature eigenvalues belong to the continuous Jacobi sector. The physical root density approaches the exact reference as D grows. At $D = 546$, the rank first exceeds the one-polarization capacity: Eq. (9) forces six eigenvalues at each boundary. At $(D, M) = (800, 680)$, there are 120 atoms per boundary, or 30.0% of all eigenvalues.

The atom-free interior remains chaotic on both sides

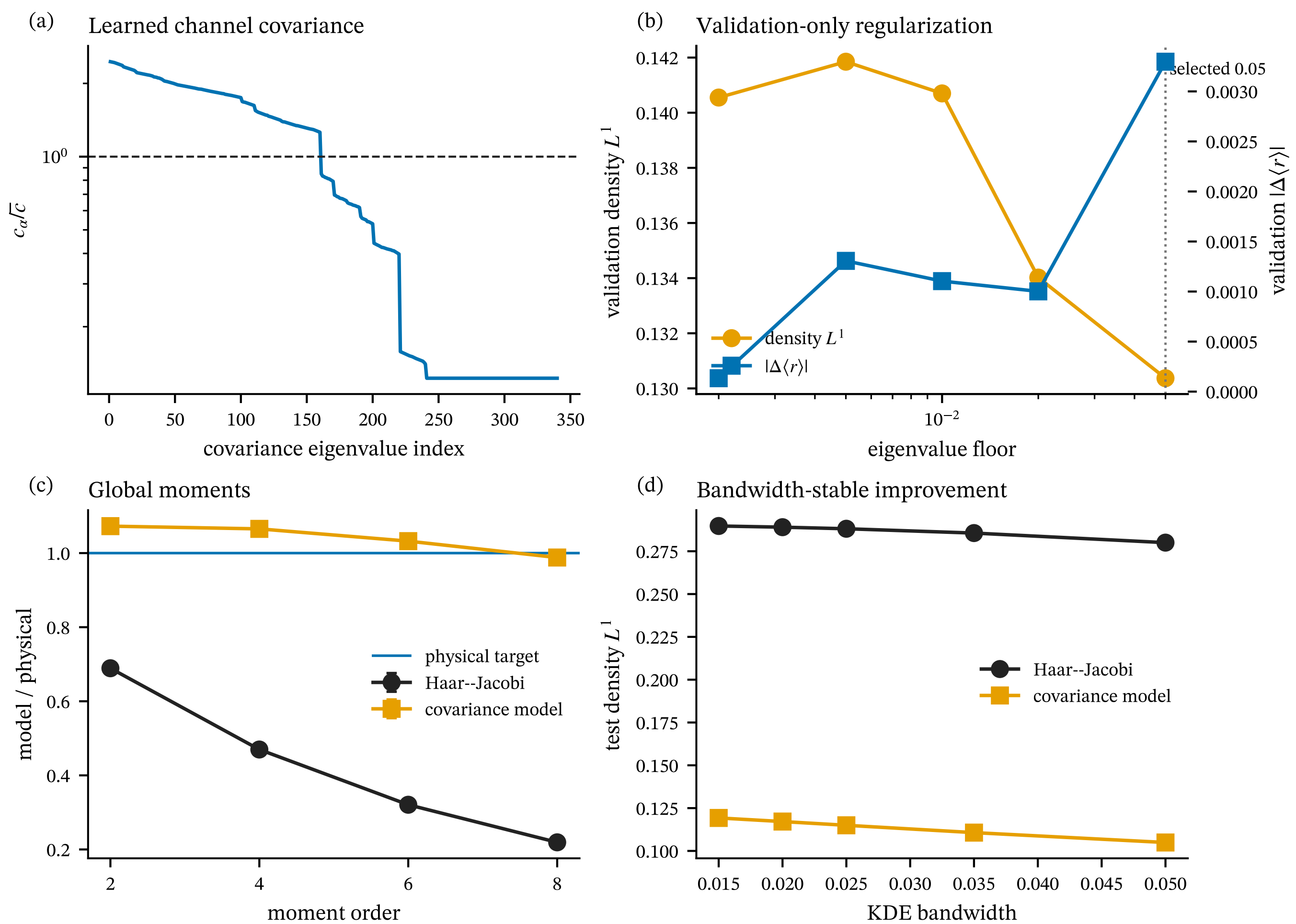

FIG. 2. **Covariance mechanism and held-out model selection.** (a) Eigenvalues of the learned regularized channel covariance, normalized by their mean; the Haar value is unity. (b) Validation density and local-gap errors for the five candidate eigenvalue floors. The vertical line denotes the validation-selected floor. (c) Model moments divided by the physical target. Covariance deformation captures most of the global moment hierarchy, whereas Haar increasingly underestimates the tails. (d) Test-set density error under KDE bandwidth variation.

of the transition. At the largest point its density error is 0.0523, while the adjacent-gap difference remains at the few-10^{-3} scale. The exact modes therefore do not "poison" the complementary spectrum. They record a deterministic intersection of the active row space with the two polarizations of J, while the generic orientation of the remaining subspace supplies Jacobi repulsion.

This decomposition is directly relevant to degenerate quantum matter. Exact curvature eigenvectors may be imposed by accessibility, selection rules, or topology even when the accessible complement is geometrically chaotic. A diagnostic based only on the total one-point density could misclassify such a system because the delta functions dominate high moments. The correct procedure is to identify exact subspaces algebraically and test randomness on the remaining support.

C. Finite-size evidence

Figure 4 compares three deviations from the Haar–Jacobi null model. Errors are obtained from seed-block fluctuations. For each observable we fit nonnegative-offset forms

$$y(D) = y_\infty + AD^{-p} \tag{17}$$

with fixed $p = 1/2$, fixed $p = 1$, and a free exponent; models are ranked by leave-one-size-out prediction.

The free density fit gives $p = 0.593$, the local gap difference favors D^{-1}, and the participation deficit favors $D^{-1/2}$. The exponents should not be read as universal critical indices. There are only seven sizes, the last two cross a kinematic boundary, and their sample counts are deliberately reduced to keep the eigenvalue budget controlled. What is robust is the directional statement: local and global discrepancies fall substantially as coordinate

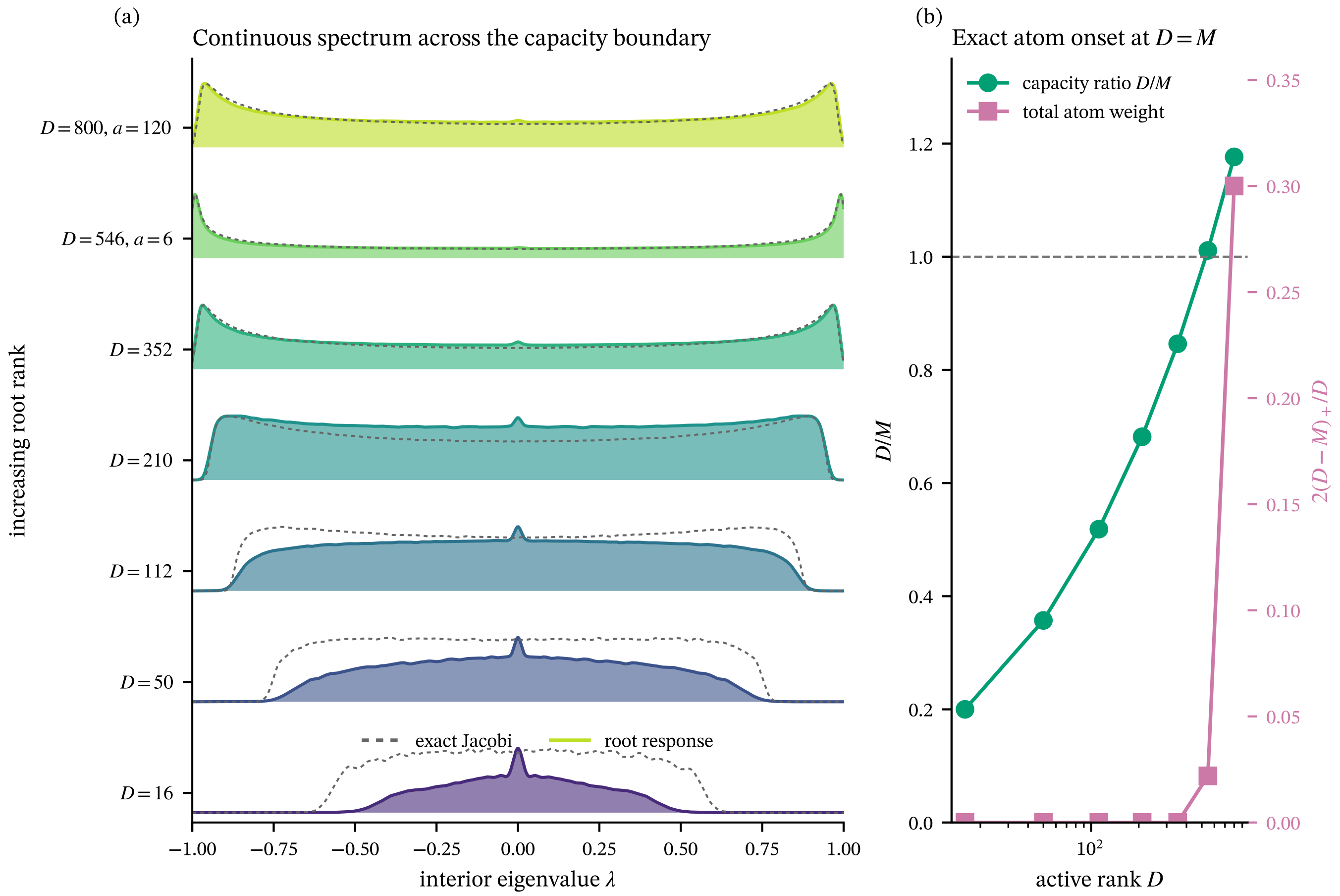

FIG. 3. **Continuous random-matrix spectrum across an exact atom transition.** (a) Ridgeline densities of the algebraically stripped interior spectrum (colored) and exact Jacobi reference (dashed) from $D = 16$ to $D = 800$. Labels a give the exact multiplicity at each boundary. (b) Capacity ratio D/M and total atom weight $2(D - M)_+/D$. The transition occurs at $D = M$.

TABLE I. **Increasing-rank root-response ensembles.** "Atoms/edge" is the algebraic multiplicity at each of $\lambda = \pm 1$; all density and gap statistics in the last two columns use only the continuously distributed interior.

D	M	samples	atoms/edge	$\Delta\langle r\rangle$	L^1_{int}
16	80	2000	0	0.03023	0.43375
50	140	2000	0	0.00767	0.22916
112	216	2000	0	0.00403	0.14094
210	308	1000	0	0.00173	0.09624
352	416	1000	0	0.00290	0.06866
546	540	500	6	0.00024	0.05149
800	680	250	120	0.00423	0.05226

participation rises, and the continuous-sector agreement survives the atom onset.

VI. LOCAL-TO-GLOBAL HIERARCHY

Random-matrix agreement is scale dependent even in conventional quantum chaos. To determine which part of the geometric spectrum is universal, we compare four statistics after ensemble-CDF unfolding: the full adjacent-gap-ratio distribution $P(r)$, the number variance $\Sigma^2(L)$, the Dyson–Mehta rigidity $\Delta_3(L)$, and the connected form factor

$$K_c(\tau) = \frac{1}{D}\left[\left\langle |Z(\tau)|^2\right\rangle - |\langle Z(\tau)\rangle|^2\right], \quad Z(\tau) = \sum_a e^{-2\pi i\tau\xi_a}. \tag{18}$$

Here ξ_a are unfolded curvature eigenvalues.

Figure 5(a) makes the strongest local statement of the paper. The three $P(r)$ curves lie within the simultaneous bands over the entire interval, not only at the mean quoted in Eq. (13). Likewise, the connected form-factor ramp is

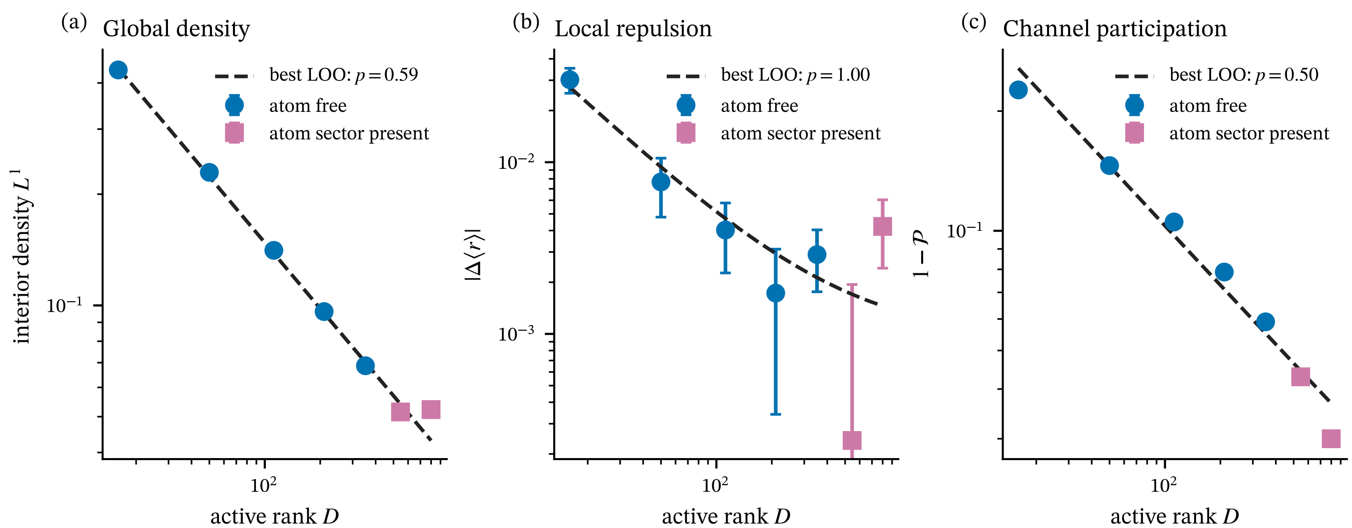

FIG. 4. **Finite-size approach to geometric universality.** Seed-block uncertainties and leave-one-size-out-selected fits for (a) interior density distance, (b) adjacent-gap-ratio difference, and (c) channel-participation deficit. Squares denote ranks with an exact atom sector. Fits describe the finite registered sequence and are not thermodynamic-limit theorems.

already close: at $\tau = 0.5$,

$$K_{c,\mathrm{phys}} = 0.502, \quad K_{c,\mathrm{Haar}} = 0.495, \quad K_{c,C} = 0.502. \tag{19}$$

The plateau is reached near $\tau = 1$ in all three ensembles.

The number variance resolves a later scale. At the longest displayed window,

$$\Sigma^2_{\mathrm{phys}}(8) = 0.696, \quad \Sigma^2_{\mathrm{Haar}}(8) = 0.570, \quad \Sigma^2_C(8) = 0.570. \tag{20}$$

The physical excess grows gradually with L, even though the rigidity estimator and form-factor ramp remain close over the accessible range. This is residual higher-order memory not captured by the learned one-body covariance.

These observations suggest a geometric analogue of a Thouless hierarchy. There is no real-time diffusion process in the curvature matrix itself, so we do not identify a literal Thouless time. Instead, the hierarchy is among resolution scales in the unfolded curvature spectrum: nearest-neighbor repulsion is universal first, the ramp follows, and broad counting windows retain microscopic memory longest. The terminology "geometric ETH" should be understood in this operational sense.

The conclusion is also more precise than declaring the physical matrices "random" or "nonrandom." They are random-matrix-like under local probes, covariance deformed under global one-point probes, and non-Haar under selected long-range and frame observables. All three statements refer to the same exactly degenerate manifold and are mutually consistent.

VII. DISCUSSION AND OUTLOOK

We have tested the Berry-curvature formulation of chaos in a fractional topological zero-mode manifold at a scale where confidence bands, held-out modeling, long-range statistics, and an exact rank transition can be resolved simultaneously. The result is not full Haar randomness. It is a hierarchy with a controlled mechanism: local curvature levels exhibit Jacobi repulsion; a learned channel covariance explains most of the global density and moment deformation; and higher Grassmannian correlations survive at the longest available scales.

The exact atom transition adds a structural refinement. When $r > M$, the normalized curvature necessarily contains $(r - M)$ eigenvalues at each signature boundary. These levels are neither an accident of finite precision nor evidence against chaos. They are dimension-theorem modes. After their algebraic removal, the complementary spectrum remains locally universal and approaches the exact dual Jacobi density. Exact geometry and chaotic geometry can therefore occupy orthogonal parts of the same degenerate manifold.

The relation to Ref. [10] is direct but not identical. That work argues that non-Abelian Berry curvature diagnoses chaos in BPS sectors where energy levels are exactly degenerate, and contrasts random black-hole microstates with structured smooth geometries. Our calculation supplies a condensed-matter realization of the same organizing principle—quantum geometry as a chaos probe—without BPS cohomology or a gravitational dual. It also makes explicit which layer of random-matrix behavior appears first and how deterministic geometric modes should be separated from a chaotic complement.

Several extensions are now sharply posed. First, one should repeat the train–validation–test analysis along a genuine many-body thermodynamic sequence rather than the fixed-N dilution route used here. Second, the covariance ansatz should be derived from locality and clustering of tangent channels, including a prediction for

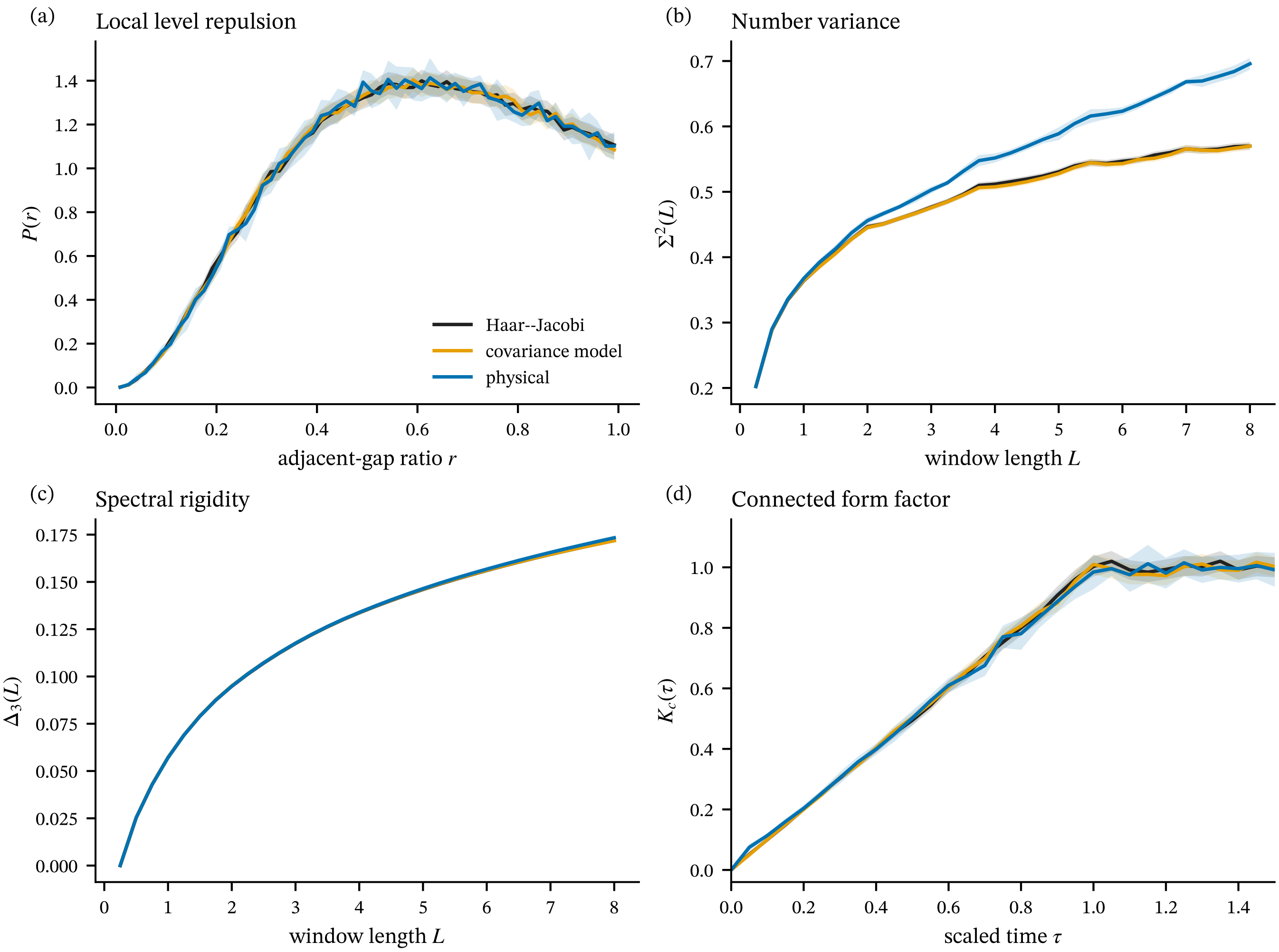

FIG. 5. **Local-to-global hierarchy of geometric chaos.** Simultaneous confidence bands for (a) the full adjacent-gap-ratio distribution, (b) number variance, (c) spectral rigidity, and (d) connected form factor. Physical uncertainties resample eight independent seed blocks; random-ensemble uncertainties resample matrices through a Gaussian multiplier bootstrap.

the residual frame cumulants. Third, the local statistics of the quantum metric and cross-correlations between metric and curvature can test whether the entire quantum geometric tensor thermalizes as one object. Fourth, integrating $\mathrm{Tr}\, F$ over a controlled two-cycle would connect the local atom/covariance structure studied here to the Chern-number topology emphasized in Ref. [10].

The practical lesson is already robust. For an exactly degenerate condensed-matter system, one should not begin by adding a splitting merely to recover conventional level statistics. One should first compute how the projector moves, normalize curvature by its metric, identify accessibility-enforced exact sectors, and then compare the continuous active spectrum with the finite-dimensional Jacobi law. This order of operations turns degeneracy from an obstruction into the defining resource of the chaos diagnostic.

Appendix A: Numerical methods and inference

1. Physical parent and tangent cache

The physical parent is built independently in this project from the magneto-periodic Kapit–Mueller hopping model and a projected on-site contact interaction [12]. The many-body parent is diagonalized once. Its zero-mode eigenvectors, complementary eigenvectors, and energy denominators define a linear cache from local-potential coefficients to the channel X_μ in Eq. (3). For every sampled potential, the cached channel is checked against its many-body Hilbert–Schmidt norm.

The physical calculation uses 20000 tangent pairs in eight seed blocks. The registered split contains training, validation, and test subsets fixed before any covariance model is evaluated. The parent kernel width is below the numerical rank tolerance and the external gap remains open. Every physical metric has active rank 50, and no boundary atoms are expected because $r < M$.

2. Root response and QR whitening

The root basis consists of cyclic $(1, 2)$-admissible occupation strings. For each one-body tangent, nonzero transitions from a root configuration to a first descendant are precomputed as sparse integer records. Sampling a new tangent changes only the record weights. This avoids rebuilding a dense many-body operator for every matrix.

Let B be the real $D \times 2M$ doubled response matrix. A thin factorization

$$B^T = QR \tag{A1}$$

gives a row isometry $Y = Q^T$ up to an active-space orthogonal transformation. Because this transformation only conjugates $YJY^\dagger$, its eigenvalues agree with explicit metric inverse-square-root normalization. Focused regression tests compare the two constructions directly.

For $r > M$, atom labels are assigned before examining floating-point eigenvalues: the first $r - M$ ordered levels are negative atoms and the last $r - M$ are positive atoms. The numerical values are then audited against ± 1. Only the remaining $2M - r$ values enter unfolding, density, and local-spacing statistics.

3. Independent Jacobi sampling

For interior dimension $k = \min(r, 2M-r)$ and exponent $a = |M-r|$, two independent $k \times (k+a) = k \times M$ complex Gaussian matrices generate Wishart matrices A and B. The generalized eigenvalues of

$$Au = t(A + B)u \tag{A2}$$

are mapped to $\lambda = 2t - 1$. This matrix-beta sampler produces the exact complex Jacobi weight in Eqs. (7) and (10). When $r > M$, the algebraic ± 1 atoms are added explicitly. Random seeds are independent of every physical and root-response seed.

4. Covariance fitting without leakage

The mean projector in Eq. (14) is accumulated from 1024 training row spaces. The raw matrix is positive semidefinite but rank deficient. For a candidate fraction f, its eigenvalues are replaced by

$$c_\alpha \longmapsto \max(c_\alpha, f c_{\max}) \tag{A3}$$

and the trace is normalized to the ambient dimension. Five values of f are evaluated with common random numbers. The validation score is the density L^1 distance plus one half of the absolute mean-gap-ratio difference. Only after selecting $f = 0.050$ do we generate the final 10000 model matrices and compare them with the untouched test set.

5. Matrix-level uncertainty

An eigenvalue is not an independent experimental unit. All uncertainty estimates therefore act on complete matrices or independent seed blocks. For the physical ensemble, a hierarchical bootstrap resamples the eight block means. For the 10000-matrix and 10000-matrix reference ensembles, we use the Gaussian multiplier limit of the matrix-mean process. Its covariance is estimated from between-matrix fluctuations of the entire curve. 10000 replicates determine both pointwise intervals and a simultaneous critical value from the maximum standardized deviation over the plotted grid.

The density uses a Gaussian KDE of registered bandwidth. The adjacent-gap distribution is computed per matrix in the central bulk. Unfolding uses the pooled ensemble CDF. Number variance is evaluated from sliding windows within each unfolded spectrum. The rigidity follows from the stationary identity

$$\Delta_3(L) = \frac{2}{L^4} \int_0^L \left(L^3 - 2L^2 x + x^3\right) \Sigma^2(x)\, \mathrm{d}x. \tag{A4}$$

The connected form factor is defined in Eq. (18).

6. Sensitivity and finite-size fits

The covariance-versus-Haar density ordering is repeated over five KDE bandwidths. The local conclusion is repeated over five bulk fractions, ensemble-CDF and polynomial unfolding, and three adjacent-gap histogram resolutions. Every registered choice preserves the qualitative conclusion.

For the rank sequence, density distance, gap-ratio difference, and participation are first computed in each of eight seed blocks. Their standard errors enter the bounded fits of Eq. (17). The fixed-1/2, fixed-1, and free-exponent models are compared by leave-one-size-out root-mean-square prediction error. The fit range includes both sides of the atom transition; no thermodynamic-limit claim is inferred from these seven points.

7. Reproducibility gates

The delivery pipeline fails if any registered sample count is reduced, the $D = 800$ case is absent, the atom counts disagree with Eq. (9), physical/validation/test indices overlap, a confidence claim pools eigenvalues, generated figures are out of sync with their input hashes, LaTeX references are undefined, or a PDF page is not rendered for visual inspection. The task-local source imports no executable or artifact from the preceding manuscript task.

ACKNOWLEDGMENTS

The numerical artifacts, fixed train–validation–test split, matrix-level uncertainty estimates, and figure-generation scripts are retained with the source of this manuscript.

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